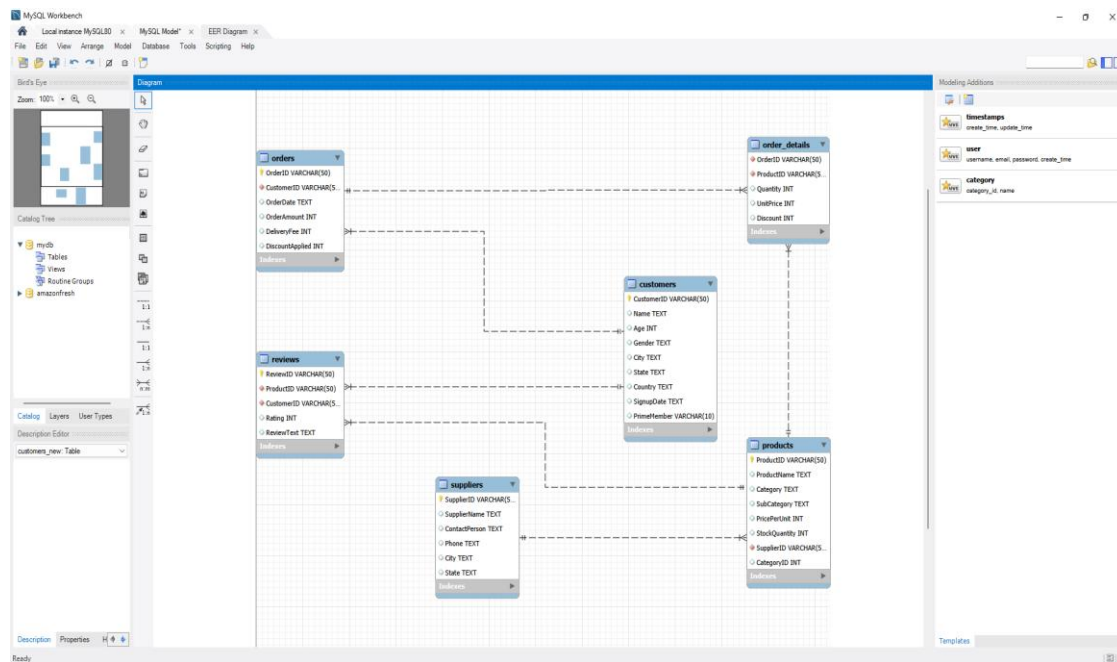


Amazon Fresh Analytics Project Documentation

Task 1 - ER Diagram

An Entity Relationship Diagram (ERD) was created to represent the structure of the Amazon Fresh database. The diagram illustrates the relationships between Customers, Orders, Order Details, Products, Suppliers, Reviews, and Product Categories tables.

Primary keys and foreign keys were used to establish relationships and maintain referential integrity.



Task 2 - Primary Keys and Foreign Keys

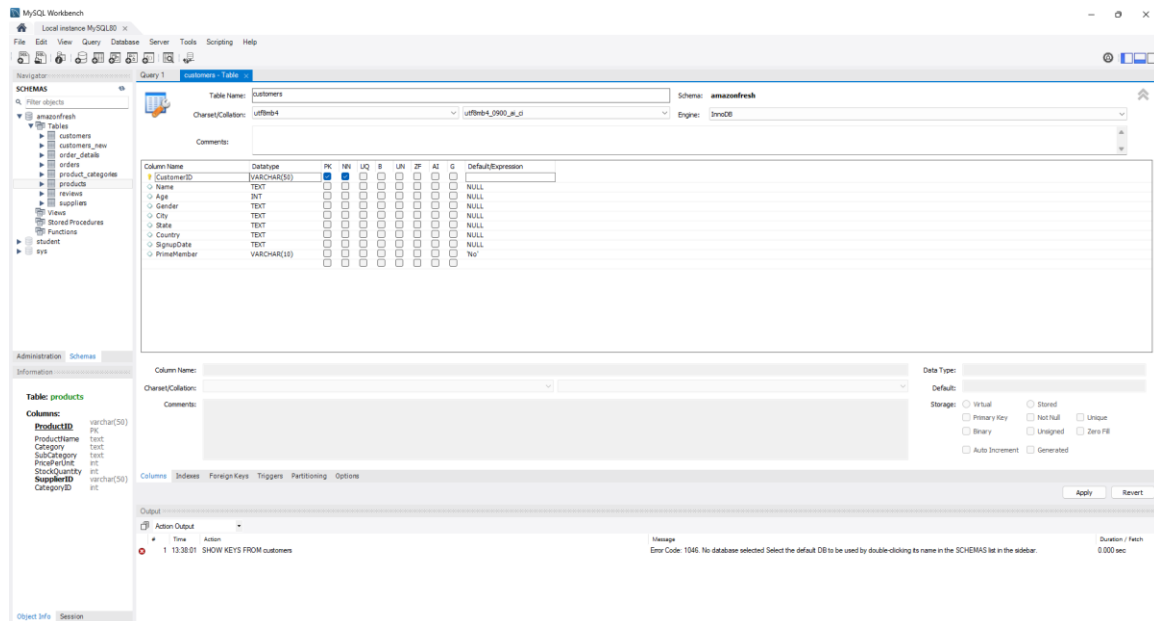
Primary keys were created for all major tables to uniquely identify records. Foreign keys were added to establish relationships between related tables.

Relationships implemented:

- Customers → Orders
- Orders → Order Details
- Products → Order Details
- Suppliers → Products
- Customers → Reviews

- Products → Reviews

These relationships ensure data consistency and integrity.



Task 3 - Basic Queries

Basic SQL queries were executed to retrieve information from the database.

Operations performed:

- Display customer records
- Display product records
- Retrieve specific information using SELECT statements
- Filter records using WHERE conditions

These queries helped understand the structure and content of the database.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

Filter objects

ascascfresh

Tables

customers

order_details

orders

products

reviews

suppliers

Views

Stored Procedures

Functions

Tables

Information

Table: products

Columns:

ProductID varchar(30)

ProductName text

Category text

SubCategory text

StockQuantity int

SupplierID int

Query 1

customers - Table order_details - Table orders - Table order_details - Table customers - Table

Limit to 1000 rows

143 AND REFERENCED_TABLE_NAME IS NOT NULL.)

144 SELECT *

145 FROM customers

146 WHERE City = 'Pattberg'

147 SELECT *

Result Grid

ProductID	ProductName	Category	SubCategory	StockQuantity	SupplierID
1043000-848F-4368-477a-2053263-040	Apple Fruit	Fruit	Sub-Fruit-4	110	295
0442007-4242-4058-6276-1273739-0404	Apple Fruit	Fruit	Sub-Fruit-2	887	32
0786236-434F-4338-8445-4826850-0807	Community Fruit	Fruit	Sub-Fruit-3	419	461
0419000-8604-4251-9405-4804807-7726	Cash Fruit	Fruit	Sub-Fruit-4	515	209
0276273-6207-4376-4a52-74632630-0404	Cash Fruit	Fruit	Sub-Fruit-2	655	467
0616403-1830-86a3-9763-aa38889-7746	Flat Fruit	Fruit	Sub-Fruit-2	976	383
0786866-6666-4464-6a77-8888a-104a7a	Imperial Fruit	Fruit	Sub-Fruit-3	492	352
1334774-c68e-438F-4145-705a633-139	Blue Fruit	Fruit	Sub-Fruit-2	8	210
1453236-4250-1451-1678-6aa3200-6a87	Fruit Fruit	Fruit	Sub-Fruit-4	793	460
17593a76-c442-4323-8046-675a-4533-79e	Wild Fruit	Fruit	Sub-Fruit-1	830	231
1231348-673e-4a48-a676-ba-313237-79e	Soft Fruit	Fruit	Sub-Fruit-4	122	39
3844867-ae57-4553-4557-68809e9-7679	Repeat Fruit	Fruit	Sub-Fruit-1	641	71
20336c3-5227-1037-4033-4a04747-7974	Poplar Fruit	Fruit	Sub-Fruit-1	94	256
24638276-c513-4323-aa32-8a2c3a2f-7a0b	Hemlock Fruit	Fruit	Sub-Fruit-1	207	266
2ae3875-f3a1-4253-869a-46a42c3-79a	Cash Fruit	Fruit	Sub-Fruit-4	26	336
3586274-648e-86c4-6455-4215032-3a0d	Cash Fruit	Fruit	Sub-Fruit-1	366	229
3274966-05a4-46a6-9a02-475a6c9-8e1c3	Belene Fruit	Fruit	Sub-Fruit-1	745	326
3a2004d-f6b1-46b1-80a4-47946222-32c1	Pony Fruit	Fruit	Sub-Fruit-3	504	270
3a81460-3b45-44e6-9a25-0a35498-02f	Lost Fruit	Fruit	Sub-Fruit-2	122	479
390348d-4043-a43a-6232-64357-1a3002	Wine Fruit	Fruit	Sub-Fruit-4	247	37
40e128a-17d7-41a9-4311-4498323-3003	Situation Fruit	Fruit	Sub-Fruit-2	404	66
4e76622-05a6-46a6-83a6-53a-1237-9e	Unit Fruit	Fruit	Sub-Fruit-2	404	329
4a3a047a76-74fe-69ff-363a580-9a103	Example Fruit	Fruit	Sub-Fruit-2	880	260
4b30a68-478a-4048-a3a8-9a3a-135a-7a0d	Institution Fruit	Fruit	Sub-Fruit-4	334	210

Output

Action Output

Time Action

114 12:39:11 ALTER TABLE order_details ADD CONSTRAINT fk_orderdetails_products FOREIGN KEY (ProductID) REFERENCES products(ProductID) 10000 rows affected Records: 10000 Duplicates: 0 Warnings: 0 0.312 sec

116 12:39:36 SELECT TABLE_NAME, COLUMN_NAME, CONSTRAINT_NAME, REFERENCED_TABLE_NAME, REFERENCED_COLUMN_NAME FROM information.

117 12:40:35 SELECT * FROM customers WHERE City = 'Pattberg' LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec

118 12:40:55 SELECT * FROM products WHERE Category = 'Fruit' LIMIT 0, 1000 89 row(s) returned 0.000 sec / 0.000 sec

119 12:41:10 SELECT * FROM customers WHERE City = 'Pattberg' LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec

120 12:42:05 SELECT * FROM products WHERE Category = 'Fruit' LIMIT 0, 1000 89 row(s) returned 0.000 sec / 0.000 sec

Query Completed

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

Filter objects

ascascfresh

Tables

customers

order_details

orders

products

reviews

suppliers

Views

Stored Procedures

Functions

Tables

Information

Table: products

Columns:

ProductID varchar(30)

ProductName text

Category text

SubCategory text

StockQuantity int

SupplierID int

Query 1

customers - Table order_details - Table orders - Table order_details - Table customers - Table

Limit to 1000 rows

142 SELECT

143 TABLE_NAME, COLUMN_NAME, CONSTRAINT_NAME, REFERENCED_TABLE_NAME, REFERENCED_COLUMN_NAME

144 FROM information_schema.KEY_COLUMN_USAGE

145 WHERE TABLE_SCHEMA = 'ascascfresh'

146 AND REFERENCED_TABLE_NAME IS NOT NULL.)

147 SELECT *

148 SELECT *

Result Grid

CustomerID	Name	Age	Gender	City	Date	Country	Signature	ProfilePhoto
432159f6-04-0483-62b1-43994-795a	Larry Gallagher	54	Male	Pattberg	2023-09-07	India		
0209		0209	0209	0209	0209	0209		

Output

Action Output

Time Action

114 12:38:59 ALTER TABLE order_details ADD CONSTRAINT fk_orderdetails_orders FOREIGN KEY (OrderID) REFERENCES orders(OrderID) 10000 rows affected Records: 10000 Duplicates: 0 Warnings: 0 0.281 sec

116 12:39:11 ALTER TABLE order_details ADD CONSTRAINT fk_orderdetails_products FOREIGN KEY (ProductID) REFERENCES products(ProductID) 10000 rows affected Records: 10000 Duplicates: 0 Warnings: 0 0.312 sec

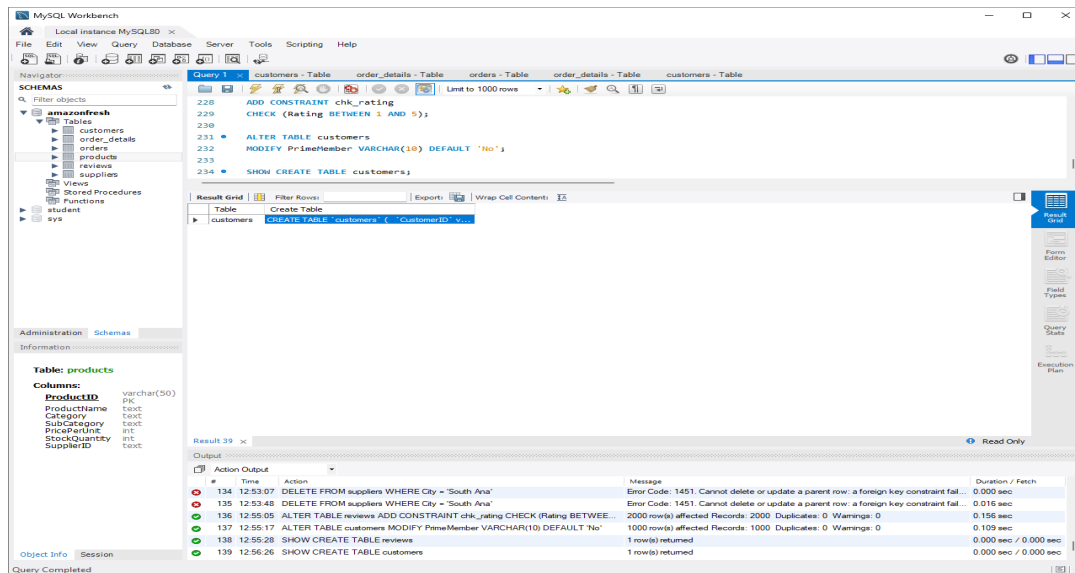
117 12:39:36 SELECT TABLE_NAME, COLUMN_NAME, CONSTRAINT_NAME, REFERENCED_TABLE_NAME, REFERENCED_COLUMN_NAME FROM information.

118 12:40:35 SELECT * FROM customers WHERE City = 'Pattberg' LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec

119 12:40:55 SELECT * FROM products WHERE Category = 'Fruit' LIMIT 0, 1000 89 row(s) returned 0.000 sec / 0.000 sec

120 12:41:10 SELECT * FROM customers WHERE City = 'Pattberg' LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec

Query Completed



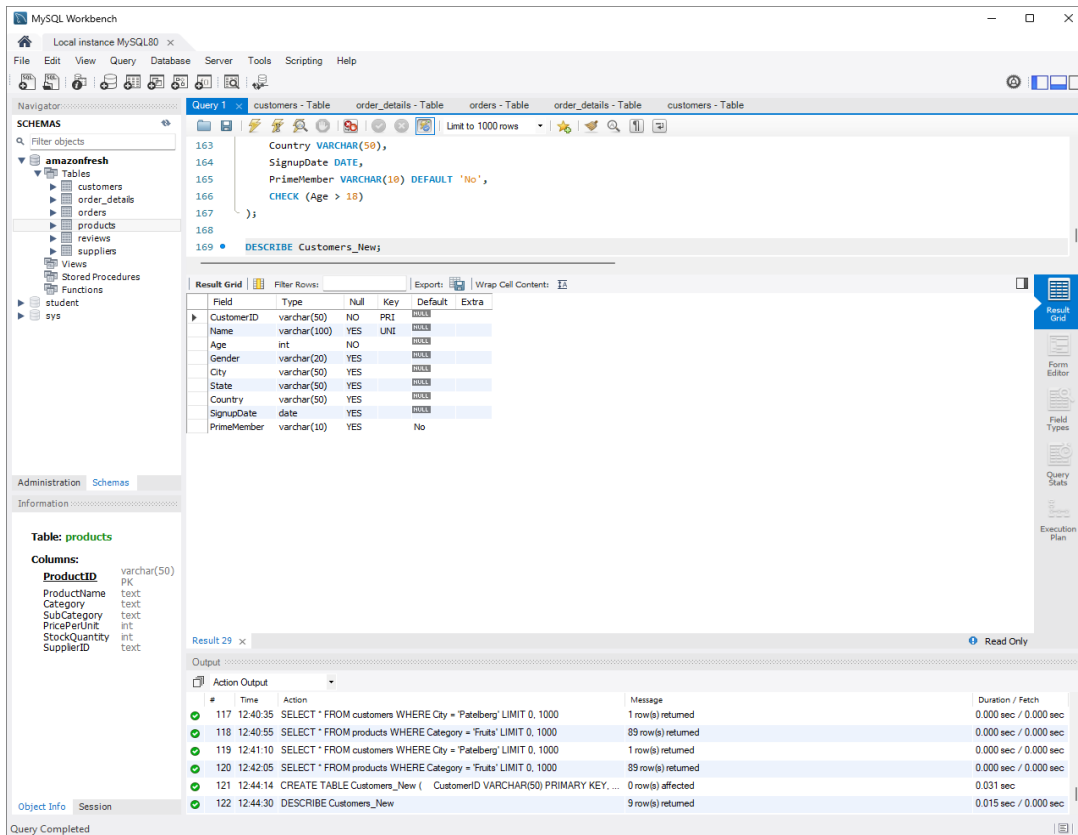
Task 4 - DDL Operations

Data Definition Language (DDL) commands were used to create and modify database objects.

Operations performed:

- CREATE TABLE
- ALTER TABLE
- MODIFY COLUMN
- ADD CONSTRAINT

DDL commands define the structure of the database and ensure proper table design.



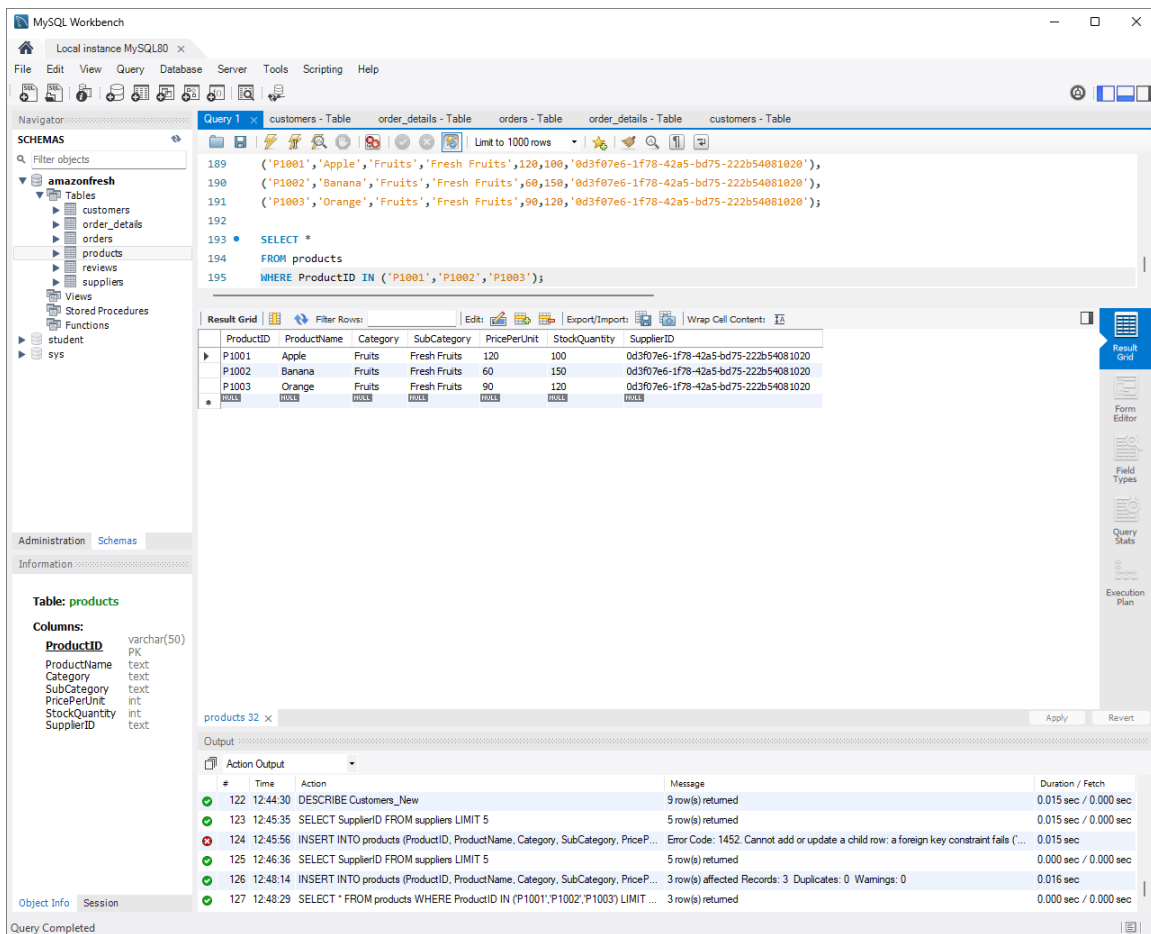
Task 5 - INSERT Operations

New product records were inserted into the Products table.

Sample products inserted:

- Apple
- Banana
- Orange

The INSERT statement was used to add new rows while maintaining data integrity through foreign key relationships.



Task 6 - UPDATE Operations

The UPDATE statement was used to modify existing records.

Example:

- Updated stock quantity of Product ID P1001.

The UPDATE command allows modification of existing information without affecting other records.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Filter objects

SCHEMAS

- amazonfresh
 - Tables
 - customers
 - order_details
 - orders
 - products
 - reviews
 - suppliers
 - Views
 - Stored Procedures
 - Functions
 - student
 - sys

Administration Schemas

Information

Table: products

Columns:

- ProductID: varchar(50), PK
- ProductName: text
- Category: text
- SubCategory: text
- PricePerUnit: int
- StockQuantity: int
- SupplierID: text

Query 1: customers - Table, order_details - Table, orders - Table, order_details - Table, customers - Table

```

189 ('P1001','Apple','Fruits','Fresh Fruits',120,100,'0d3f07e6-1f78-42a5-bd75-222b54081020'),
190 ('P1002','Banana','Fruits','Fresh Fruits',60,150,'0d3f07e6-1f78-42a5-bd75-222b54081020'),
191 ('P1003','Orange','Fruits','Fresh Fruits',90,120,'0d3f07e6-1f78-42a5-bd75-222b54081020');
192
193 SELECT *
194 FROM products
195 WHERE ProductID IN ('P1001','P1002','P1003');
  
```

Result Grid

ProductID	ProductName	Category	SubCategory	PricePerUnit	StockQuantity	SupplierID
P1001	Apple	Fruits	Fresh Fruits	120	100	0d3f07e6-1f78-42a5-bd75-222b54081020
P1002	Banana	Fruits	Fresh Fruits	60	150	0d3f07e6-1f78-42a5-bd75-222b54081020
P1003	Orange	Fruits	Fresh Fruits	90	120	0d3f07e6-1f78-42a5-bd75-222b54081020

products 32 x

Output

#	Time	Action	Message	Duration / Fetch
122	12:44:30	DESCRIBE Customers_New	9 row(s) returned	0.015 sec / 0.000 sec
123	12:45:35	SELECT SupplierID FROM suppliers LIMIT 5	5 row(s) returned	0.015 sec / 0.000 sec
124	12:45:56	INSERT INTO products (ProductID, ProductName, Category, SubCategory, PriceP...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (...)	0.015 sec
125	12:46:36	SELECT SupplierID FROM suppliers LIMIT 5	5 row(s) returned	0.000 sec / 0.000 sec
126	12:48:14	INSERT INTO products (ProductID, ProductName, Category, SubCategory, PriceP...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
127	12:48:29	SELECT * FROM products WHERE ProductID IN ('P1001','P1002','P1003') LIMIT ...	3 row(s) returned	0.000 sec / 0.000 sec

Query Completed

Task 7 - DELETE Operations

The DELETE statement was used to remove supplier records based on city conditions.

During execution, MySQL prevented deletion because the supplier record was referenced by products through a foreign key relationship.

This demonstrates referential integrity and data protection mechanisms.

Task 8 - Constraints

Constraints were implemented to enforce data validation rules.

Constraints added:

- CHECK constraint on Review Rating (1 to 5)
- DEFAULT constraint for PrimeMember field

These constraints improve data accuracy and consistency.

The screenshot shows the MySQL Workbench interface with a local instance of MySQL 8.0. The left sidebar displays the 'Schemas' tree, including 'amazonfresh' and its tables: 'customers', 'order_details', 'orders', 'products', 'reviews', and 'supplies'. The 'products' table is selected, showing its columns: ProductID (PK, varchar(50)), ProductName (text), Category (text), SubCategory (text), PricePerUnit (int), StockQuantity (int), and SupplierID (text).

The main window displays a SQL query editor with the following queries:

```

228 ADD CONSTRAINT chk_rating
229 CHECK (Rating BETWEEN 1 AND 5);
230
231 ALTER TABLE customers
232 MODIFY PrimeMember VARCHAR(10) DEFAULT 'No';
233
234 SHOW CREATE TABLE reviews;

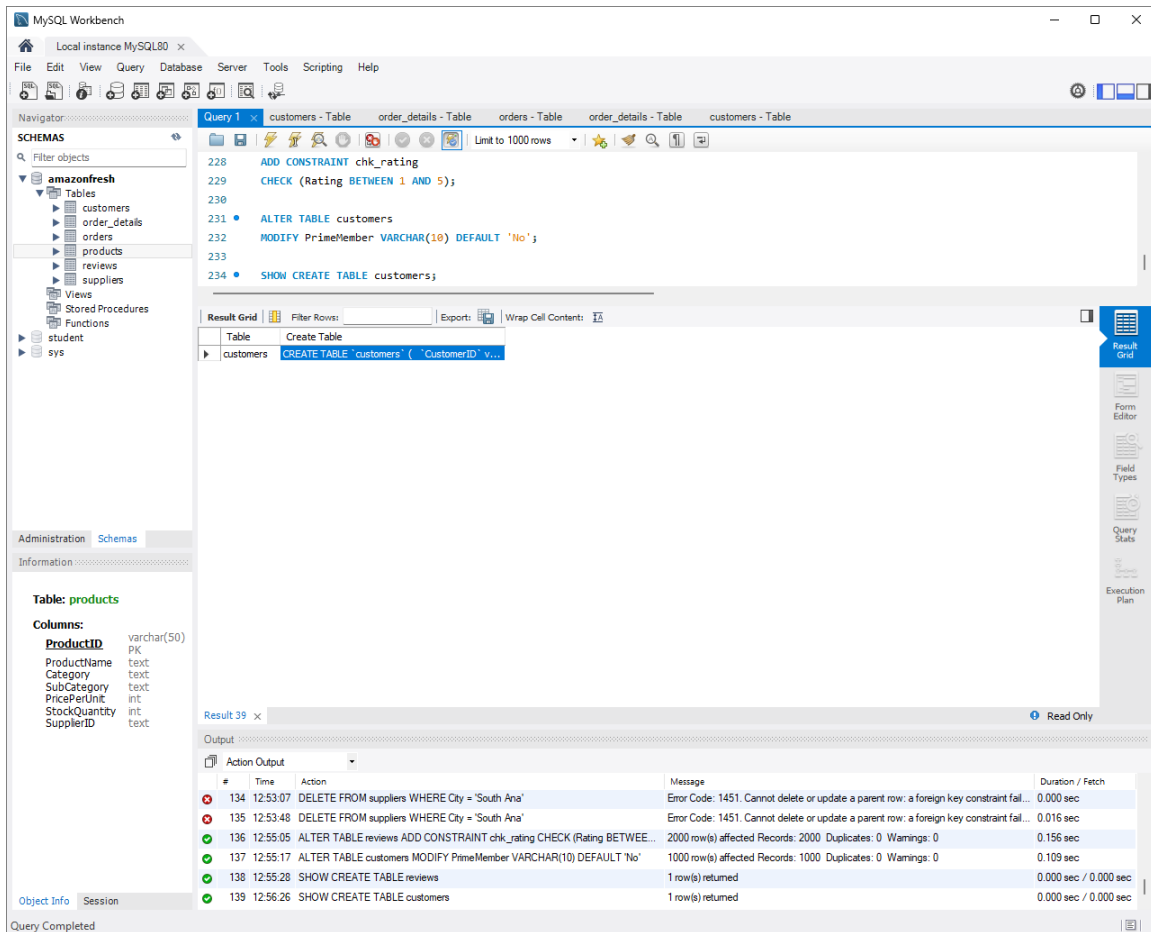
```

The 'Result Grid' shows the output of the queries, including the 'CREATE TABLE' statement for 'reviews'.

The 'Output' tab shows the execution results of the queries, including the 'ALTER TABLE' and 'SHOW CREATE TABLE' queries. The output is as follows:

#	Time	Action	Message	Duration / Fetch
133	12:52:52	SELECT * FROM suppliers WHERE City = 'South Ana' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
134	12:53:07	DELETE FROM suppliers WHERE City = 'South Ana'	Error Code: 1451. Cannot delete or update a parent row: a foreign key constraint fail...	0.000 sec
135	12:53:48	DELETE FROM suppliers WHERE City = 'South Ana'	Error Code: 1451. Cannot delete or update a parent row: a foreign key constraint fail...	0.016 sec
136	12:55:05	ALTER TABLE reviews ADD CONSTRAINT chk_rating CHECK (Rating BETWEEN 1 AND 5);	2000 row(s) affected Records: 2000 Duplicates: 0 Warnings: 0	0.156 sec
137	12:55:17	ALTER TABLE customers MODIFY PrimeMember VARCHAR(10) DEFAULT 'No'	1000 row(s) affected Records: 1000 Duplicates: 0 Warnings: 0	0.109 sec
138	12:55:28	SHOW CREATE TABLE reviews	1 row(s) returned	0.000 sec / 0.000 sec

The 'Query Completed' status is shown at the bottom of the window.



Task 9 - GROUP BY, HAVING, ORDER BY

Aggregate functions and grouping operations were performed.

Operations:

- Group suppliers by city
- Count suppliers in each city
- Sort results using ORDER BY
- Filter grouped results using HAVING

These operations help summarize and analyze data effectively.

MySQL Workbench interface showing a query window with SQL code and a result grid.

Query 1:

```
ALTER TABLE customers
MODIFY PrimeMember VARCHAR(10) DEFAULT 'No';

SHOW CREATE TABLE customers;

SELECT
```

Result Grid:

OrderID	CustomerID	OrderDate	OrderAmount	DeliveryFee	DiscountApplied
0000377-142c-4349-a419-51a985c4921b	20a9c235-9f21-468e-b261-7779f4229992	2025-01-01	709	20.1	52
00038687-999e-4887-93f5-e4311c44582a	a78d0135-3d23-4837-80f6-20f8f80f4311	2025-01-01	822	348	78
001432c4-ee5f-4e55-8868-fb3f035a804a	d02e0b75-bd26-497e-ae30-35067512f01d	2025-01-01	1545	705	61
00198b18-9c7d-ae85-9ae1-fc131b2e18815	36e69c93-7869-4d9f-bc93-19e83a8f9192	2025-01-01	1079	995	18
0015ae7a-d014-abae-979e-408e1f809297	91ba5b6d-567a-4c0e-af0c-d113ee18f2a3	2025-01-01	344	336	6
003a7963-199e-4aa2-b1a0-426a59e180e2	ca8ab063-c586-kcc5-9a0a-ebd3a8c46044	2025-01-01	6228	475	55
004ba700-015a-4107-b428-44d4-4ae213192	728328ff-f357-4849-871a-6aaf95a17a50	2025-01-01	7222	702	45
004cee1f-cf4b-4f90-a54d-68a3a2f85174	8456bc0f-690a-403a-b14e-16c375a0005	2025-01-01	9205	5	23
005acc15-a90f-4465-a342-41a4367187	20d88bd7-80f7-48b8-9b14-22269082d529	2025-01-01	5536	773	54
0068f015-c15e-ae72-918e-c2247f46c8cb	7748203c-0ea9-4603-bd6a-528c247d93f8	2025-01-01	9576	216	91
0087548e-23d6-46ed-855d-1d09144e49e4	656ba777-d04d-49bf-afc3-dfd1d428b57a	2025-01-01	8187	579	51
00ed114c-166f-416a-a1f6-914b1b701abf	518ce431-0a92-4b5c-90ff-a28fccc3e623	2025-01-01	9914	892	55
00bde197-aeaf-4d33-8bd5-240097128e61	a750a23b-83b6-43c4-9566-09e8e4e91888e	2025-01-01	2777	182	77
00c161d5-9e0b-4b65-927a-6ecf4331f532	74dae471-3fc2-4c0d-a83f13b0a18667	2025-01-01	1725	833	85
00c21d5e-6dfe-4e9c-af80-473c071f62f3	82ea7efb-6d18-42d2-a9c3-9e2f4818e6d9	2025-01-01	4231	833	11
00c49830-603a-438c-8834-ee9ec6919322	bfb282d1-a893-46f6-b941-09d81fe57585	2025-01-01	8509	25	52
00da5d80-ae3b-4e39-b7ae-2a819f676ffe	f494af69-ae92-4b02-87c8-fc8c4bcb8a3a	2025-01-01	8305	200	66
00e7678b-55cf-4a04-8823-61c70a8a1863	5cc09399-4300-4bd3-bd18-382ba1d1980c	2025-01-01	1087	881	53
010128d9-7fbc-405d-817d-a5ab6b284972	a82c0d72-bc46-4bb4-bae2-401a37d606bd	2025-01-01	577	197	14
0103224b-a5cd-4ffa-b0ff-8e6daa0a01ba	263b6cf-9aee-aae1-b9bd-e563cf41b7c7	2025-01-01	4838	10	24
01048020-7a14-4799-941f-6a732f43b0c0	e0f3d314-daba-4421-92b4-30a0701a8b31	2025-01-01	5326	491	77
011f0c2c-aa2d-4e58-9e3b-2135ab733ee4	c249e8b4-2893-4265-adf1-2de4c4ec4911	2025-01-01	1470	785	76
01181a62-40fc-40dc-ad9b-96d9f6f6f6c4	204d314b-4098-42c5-a18a-3af862326711	2025-01-01	6436	826	56
011a4e7a-4a0a-404e-10-4a0a-404e-10-4a0a	011a4e7a-4a0a-404e-10-4a0a-404e-10-4a0a	2025-01-01	9796	777	49

Output:

#	Time	Action	Message	Duration / Fetch
137	12:55:17	ALTER TABLE customers MODIFY PrimeMember VARCHAR(10) DEFAULT 'No'	1000 row(s) affected Records: 1000 Duplicates: 0 Warnings: 0	0.109 sec
138	12:55:28	SHOW CREATE TABLE reviews	1 row(s) returned	0.000 sec / 0.000 sec
139	12:56:26	SHOW CREATE TABLE customers	1 row(s) returned	0.000 sec / 0.000 sec
140	12:57:55	SELECT * FROM orders WHERE OrderDate > 2024-01-01 LIMIT 0, 1000	1000 row(s) returned	0.015 sec / 0.000 sec
141	12:58:20	SELECT p.ProductID, p.ProductName, AVG(r.Rating) AS AverageRating FROM orders o JOIN reviews r ON o.ProductID = r.ProductID	26 row(s) returned	0.016 sec / 0.000 sec
142	12:59:10	SELECT * FROM orders WHERE OrderDate > 2024-01-01 LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
143	12:59:57	SELECT p.ProductID, p.ProductName, SUM(od.Quantity) AS TotalQuantity FROM orders o JOIN products p ON o.ProductID = p.ProductID	500 row(s) returned	0.063 sec / 0.000 sec

MySQL Workbench interface showing a query window with SQL code and a result grid.

Query 1:

```
FROM products p
JOIN reviews r
ON p.ProductID = r.ProductID
GROUP BY p.ProductID, p.ProductName
HAVING AVG(r.Rating) > 4;

SELECT
```

Result Grid:

ProductID	ProductName	TotalQuantitySold	TotalSales
1034fb7-fdce-49e2-9230-98af72d77a15	Capital Snack	203	110875
6d26d138-47e1-4082-a20f-9e8a424ac57e	Society Vegetable	166	103430
2275964-4302-4922-9a65-73ac9e7eb39	Fish Dair	166	100727
05765892-c750-44cc-96e2-31fa53d42db2	Vie Vegetable	155	98926
b75932b6-d9a8-48cc-adc0-7f30cfc1b50	Able Mea	169	98184
80ef78ab-7c57-4e42-8b90-39166e8b8af6	Through Vegetable	167	95553
0f8f8a16-6143-4b73-9c42-d001ca73c906	Company Dair	192	94990
030f542-d5f3-4387-9654-90a0c38702c	Western Mea	184	94157
57893364-125e-4c30-82c6-ad91a0fed65b	Fall Snack	175	94040
5e10e2db-4dad-4c6b-9967-d8497939a268	Better Fruit	161	90966
620b859-61aa-4134-a546-248baf8489dc	Early Fruit	163	90360
2d5c2b8d-9eef-4e22-8860-e9771dbcc0d9	Interview Dair	143	89170
9e00ba8-e88a-4fd1-b5fc-eb6b3fec64b9	Even Vegetable	160	88905
a861f6bc-bdd4-7be-885f-511a8312949c	Total Dair	152	88764
2091292a-f660-4390-8223-266c018b3e3e	Drive Mea	174	88355
5ac10d07-2d4a-43f6-af72-062092a814e	Then Vegetable	152	86320
ef8b3541-8516-4339-af8c-6b65ed2b0e2	Together Vegetable	149	86300
bbb7b4eb-0e4b-42f6-b02e-ce03d3924454	Always Snack	148	86232
a85789a-2041-4789-aabb-1d7929230b94	Early Mea	139	86195
e450d38-e08e-489f-998e-88051759fc89	So Snack	162	85831
8aba364b-d0fa-4752-9809-4987289105a	Field Fruit	146	85640
062899d-2f51-46a8-ab5e-7e54a383ca44	Late Vegetable	160	85583
63f2ab66-accd-4958-a654-7519c5114c39	Challenge Baker	157	85377
1a1f4a0a-6a07-4a0a-805a-4a0a-1a1f4a0a	Waste Snack	144	84007

Output:

#	Time	Action	Message	Duration / Fetch
138	12:55:28	SHOW CREATE TABLE reviews	1 row(s) returned	0.000 sec / 0.000 sec
139	12:56:26	SHOW CREATE TABLE customers	1 row(s) returned	0.000 sec / 0.000 sec
140	12:57:55	SELECT * FROM orders WHERE OrderDate > 2024-01-01 LIMIT 0, 1000	1000 row(s) returned	0.015 sec / 0.000 sec
141	12:58:20	SELECT p.ProductID, p.ProductName, AVG(r.Rating) AS AverageRating FROM orders o JOIN reviews r ON o.ProductID = r.ProductID	26 row(s) returned	0.016 sec / 0.000 sec
142	12:59:10	SELECT * FROM orders WHERE OrderDate > 2024-01-01 LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
143	12:59:57	SELECT p.ProductID, p.ProductName, SUM(od.Quantity) AS TotalQuantity FROM orders o JOIN products p ON o.ProductID = p.ProductID	500 row(s) returned	0.063 sec / 0.000 sec

The screenshot shows the MySQL Workbench interface. The 'Schemas' pane on the left displays the 'amazonfresh' database with tables 'customers', 'order_details', 'orders', 'products', 'reviews', and 'supplies'. The 'Query' pane shows a SQL query that calculates the average rating for each product. The 'Result Grid' displays the output of the query, listing product IDs, product names, and their average ratings. The 'Output' pane at the bottom shows the execution log with timestamps and messages for each query step.

Query 1:

```

AVG(r.Rating) AS AverageRating
FROM products p
JOIN reviews r
ON p.ProductID = r.ProductID
GROUP BY p.ProductID, p.ProductName
HAVING AVG(r.Rating) > 4;

```

Result Grid:

ProductID	ProductName	AverageRating
0f6998e-6d6e-4494-bd75-88bac4160a7e	Important Fruit	4.5000
11d08ec-ef6f-43d0-abaa-414a9c336956	Door Vegetable	4.2000
141239de-d250-4d51-b078-aaa3090c0db7	Four Fruit	5.0000
226f945-a880-4e03-aabc-35e691521d6d	Social Vegetable	4.3333
2443e18-33af-4450-bab6-7accd87e424	Stand Vegetable	4.3333
3233bd0-ced5-42e6-94c3-70502e4f08c8	Close Vegetable	4.5000
3912a751-a562-4541-a3b6-afe200950edd	Big Mea	4.3333
46678d62-8c3f-4827-a27b-f7565f4c3e51	Why Vegetable	4.5714
46c2e05c-1974-4783-90b9-a5b3885ce53e	Relate Snack	4.3333
54f7a232-20e5-46a4-abdf-d5e472835bae	Tough Fruit	4.7500
55a3d70a-c3c4-4a51-af63-ca3cc04eb799	Occur Fruit	5.0000
63b5f8f4-6a55-42ad-98d4-bd6c3785d184	Sing Fruit	4.3333
69a3595f-4a6d-4b15-9d11-728ac6859b2	Style Dair	4.1429
79aafefe-36bf-43f2-8278-c795a1dc37eb	Not Fruit	4.6667
855a6664-4a5-4696-903f-014abe1dc14c	Dark Fruit	5.0000
9a01676-59a6-41de-9269-af1f80330072	Window Vegeta...	4.4000
9b29140e-246a-47aa-af80-f6b80b3fa47	Society Fruit	5.0000
b8aced22-3877-4756-ac1e-6f078d5b8f48	Account Vegeta...	4.5000
c1948281-7c7f-49c5-98d4-e4cc0dc3882	Suddenly Snack	4.4000
d840de5c-31c2-41c9-b0de-10e84ea852a7	Big Baker	5.0000
e04c5c09-cd08-46d0-b44b-49bc4bd528c	Help Mea	5.0000
e7631f67-8576-4917-d65c-946f5518ff2a	Student Dair	4.6667
f01c510c-a8f9-4706-8e27-cc0f225a454a	Total Mea	4.5000
f28a74dd-af0c-df6-ba7a-ba7b4f7b4f7b	Ann Mea	4.0000

Output:

#	Time	Action	Message	Duration / Fetch
136	12:55:05	ALTER TABLE reviews ADD CONSTRAINT chk_rating CHECK (Rating BETWEEN 4 AND 5)	2000 row(s) affected Records: 2000 Duplicates: 0 Warnings: 0	0.156 sec
137	12:55:17	ALTER TABLE customers MODIFY NameMember VARCHAR(10) DEFAULT 'No'	1000 row(s) affected Records: 1000 Duplicates: 0 Warnings: 0	0.109 sec
138	12:55:28	SHOW CREATE TABLE reviews	1 row(s) returned	0.000 sec / 0.000 sec
139	12:56:26	SHOW CREATE TABLE customers	1 row(s) returned	0.000 sec / 0.000 sec
140	12:57:55	SELECT * FROM orders WHERE OrderDate > '2024-01-01' LIMIT 0, 1000	1000 row(s) returned	0.015 sec / 0.000 sec
141	12:58:20	SELECT p.ProductID, p.ProductName, AVG(r.Rating) AS AverageRating FROM products p JOIN reviews r ON p.ProductID = r.ProductID GROUP BY p.ProductID, p.ProductName HAVING AVG(r.Rating) > 4	26 row(s) returned	0.016 sec / 0.000 sec

Task 10 - High Value Customers

Customers with high spending behavior were identified.

Techniques used:

- JOIN between Customers and Orders
- SUM function to calculate total spending
- HAVING clause to filter high-value customers
- ORDER BY to rank customers

This analysis helps identify important customers for business growth.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'amazonfresh' database schema with tables: customers, order_details, orders, products, reviews, and supplies. The main query editor shows a SQL query that joins the 'products' and 'order_details' tables on 'ProductID' and groups the results by 'ProductID' and 'ProductName', ordered by 'TotalSales' in descending order. The 'Result Grid' shows a list of products with their respective total sales. The 'Output' pane at the bottom shows the execution log of the query.

Query 1:

```

255 FROM products p
256 JOIN order_details od
257 ON p.ProductID = od.ProductID
258 GROUP BY p.ProductID, p.ProductName
259 ORDER BY TotalSales DESC;
260
261 SELECT

```

Result Grid:

CustomerID	Name	City	TotalSpending
82ea7efb-dd18-42cd-a4e3-9e2f4519beb5	Hannah Day	New Patricia	69698
a3eb895d-40f9-423c-aabe-9b3826bdf3c	Ralph Evans	Mooreborough	69469
d736cf4e-bf76-4cc3-996c-8db1de150ed5	Yvette Murray	Robertsborough	69398
9a863734-e786-47b1-82c2-f976dbc54db3	Anthony Martinez	South Michael	67027
9dbf19e4-2148-4898-bef4-f59f65589a60	Joseph Thomas	East Loretta	66310
959d815d-bfbc-4f68-9405-f8d8eea87095	Lawrence Walsh	Mariahside	64141
f7d12d27-d34d-4871-9ebd-5d25aa83a81f	Daniel Clark	Reevesbury	62881
f4269db2-909a-444e-9d2e-fd7463b3fd	Heather Bennett	West Emily	61343
656ac2d5-21c9-42a1-8236-f2d7cb8b92e	Amanda Stevenson	Sampsonbury	60289
c20121ee-13c1-4273-92c4-1191f0fd29f8	Stacey Webb	Scottfurt	59329
b3853a75-9ce9-409b-a376-c415766a0128	Kristopher Douglas	Angelaft	59018
b7eedb50-01bb-41c6-8282-58684b0e0b51	Thomas Bond	Scotmouth	57924
3564f8a0-5021-4c80-808e-26dd1021c174	Michelle Smith	Clarkland	57528
8a399b7e-9c66-4785-99d9-e143f9f1ac8e	Michael Jackson	Whitneybury	57476
e6d6344e-debd-4788-a203-eece2a89ce3e	Holly Miller	South Stacymo...	56375
aeff7c7d-e98e-4c38-80dc-01eb98f6f096	Margaret Davis	Port Patricia	55165
0577c43a-b991-44d0-b531-e2adaf9acbf	Leslie Martinez	Lake Kennethb...	55048
6de4ff6-1573-4c85-8366-cfcc5de9fe1f	Richard Meadows	New Joseph	54887
4cafcafe-fae0-4a04-9477-4e638b76b192	Amy Davis	Priceland	54750
ba87e769-be70-4814-83d6-4701f971a7b0	Alexander Baker	New Monique	54732
6bb012e-5e78-46f6-8766-a0cf3eebd41e	Allison Carson	Ericshire	54439
d9169a3d-d07f-4edd-b1d4-bb3baa23405	Jennifer Bullock	Stevenmouth	54307
154f15eb-d4b2-4b62-8088-90920093cec	Cassandra Klein	West Kristyfurt	54153
41479daa3b-771c-487a-b308-4f9b3cfa7bf6	Aaron Duman	North Lisa	53870

Table: products

Columns:

- ProductID: varchar(50), PK
- ProductName: text
- Category: text
- SubCategory: text
- PricePerUnit: int
- StockQuantity: int
- SupplierID: text

Output:

#	Time	Action	Message	Duration / Fetch
139	12:56:26	SHOW CREATE TABLE customers	1 row(s) returned	0.000 sec / 0.000 sec
140	12:57:55	SELECT * FROM orders WHERE OrderDate > '2024-01-01' LIMIT 0, 1000	1000 row(s) returned	0.015 sec / 0.000 sec
141	12:58:20	SELECT p.ProductID, p.ProductName, AVG(rf.Rating) AS AverageRating F...	26 row(s) returned	0.016 sec / 0.000 sec
142	12:59:10	SELECT * FROM orders WHERE OrderDate > '2024-01-01' LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec
143	12:59:57	SELECT p.ProductID, p.ProductName, SUM(od.Quantity) AS TotalQuantity...	500 row(s) returned	0.063 sec / 0.000 sec
144	13:01:12	SELECT c.CustomerID, c.Name, c.City, SUM(o.OrderAmount) AS TotalS...	956 row(s) returned	0.047 sec / 0.000 sec

Task 11 - Joins and Aggregations

Various JOIN operations were used to combine data from multiple tables.

Analysis performed:

- Customer order summary
- Product sales summary
- Supplier product count

Aggregate functions such as COUNT() and SUM() were used to generate meaningful insights.

MySQL Workbench interface showing a query execution for a JOIN operation. The query is:

```
276 COUNT(o.OrderID) AS TotalOrders,
277 SUM(o.OrderAmount) AS TotalSpent,
278 FROM customers c
279 INNER JOIN orders o
280 ON c.CustomerID = o.CustomerID
281 GROUP BY c.CustomerID, c.Name
282 ORDER BY TotalSpent DESC;
```

The result grid shows columns: CustomerID, Name, TotalOrders, TotalSpent. The output shows the execution plan and results for the query.

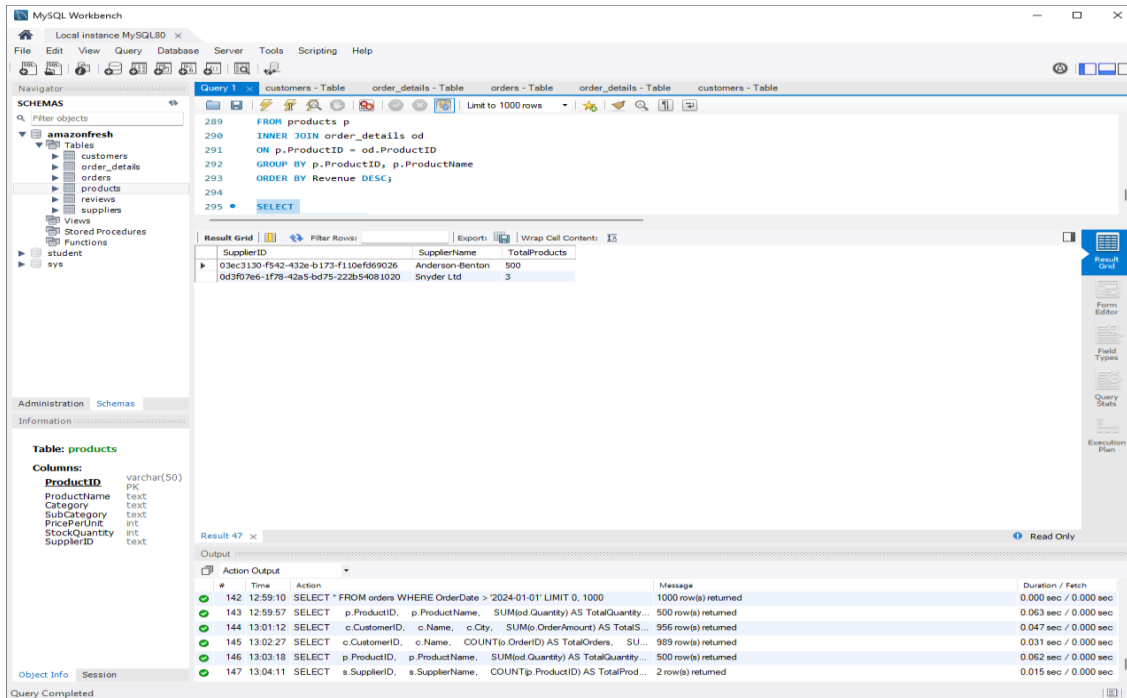
CustomerID	Name	TotalOrders	TotalSpent
82ea7efb-dd18-42cd-a4e3-9e2f4519beb5	Hannah Day	12	69608
a3ab999d-40f9-423e-aab6-4b3b26d0ff3c	Ralph Evans	10	69469
d736c4fe-b7f6-4cc3-996c-8db1de150ed5	Yvette Murray	12	69398
9a8b3734-e786-47b1-82c2-876bcb54db3	Anthony Martinez	9	67027
9dbf19e4-2148-4098-bef4-f99f5599a60	Joseph Thomas	11	66310
959d815d-bfbc-4f68-9405-f8d8ea87095	Lawrence Walsh	10	64141
7fd12d27-d34d-4b71-9ebd-5d25aab3ab1f	Daniel Clark	11	62881
f4296b2c-90ba-44e6-9d2e-7617463b3f0d	Heather Bennett	9	61343
656ac2d5-21c9-42a1-8236-f3d7fcb892e	Amanda Stevenson	9	60289
c3d121ae-13c1-4273-9a24-1194f0d29f8	Stacy White	12	59329
b3053a75-9c49-409e-a376-c41566a0128	Kristopher Douglas	9	59018
b7aeab50-01bb-41c5-8282-58684b0eb51	Thomas Bond	8	57924
356f9ba0-5021-4a30-808e-d6dd1021c174	Michelle Smith	10	57528
8a399b7e-9c66-4785-9949-e1439f1acbe	Michael Jackson	9	57476
ee6b344e-d6bd-4789-a203-aece2a890c3e	Holly Miller	12	56375
ae6ffc2d-e08e-4c38-800e-01eb08ff6d96	Margaret Davis	10	55165
0577c43a-b911-444d-b531-e2adafa9ca8f	Leslie Martinez	9	55048
6de44ff4-1573-4c85-8366-cfc0cde9e1f	Richard Meadows	9	54887
4a3cf4ee-faed-4a04-9a77-aad3bb79a102	Amy Davis	8	54750
ba87e769-be70-4b14-83d6-4701f971a760	Alexander Baker	9	54732
6dbfc12e-7846-486c-96e2-31fa1345a2b2	Allison Carson	10	54439
d916ba3d-d57f-4edd-b1d4-bb32baa23405	Jennifer Bullock	11	54307
154f15eb-d4b2-4b62-8088-90920093cec	Cassandra Klein	11	54153
41c7c7a6ab377c3a7a7a37a7a7a7a7a7a7a	Aaron B. Smith	10	63876

MySQL Workbench interface showing a query execution for a JOIN operation. The query is:

```
278 FROM customers c
279 INNER JOIN orders o
280 ON c.CustomerID = o.CustomerID
281 GROUP BY c.CustomerID, c.Name
282 ORDER BY TotalSpent DESC;
```

The result grid shows columns: ProductID, ProductName, TotalQuantitySold, Revenue. The output shows the execution plan and results for the query.

ProductID	ProductName	TotalQuantitySold	Revenue
1024867-65ea-49e2-9230-98af72d7fa15	Capital Snack	203	110875
6d26d138-47e1-4082-a20f-8ea424ac57e	Society Vegetable	166	103430
22759964-63e2-4923-9a65-73ac7e7e8b39	Fish Dair	166	100727
05765992-4780-48cc-96e2-31fa1345a2b2	Wie Vegetable	135	98926
b75932b6-49a8-48cc-edc0-7f30c6f1b50	Able Mea	169	98184
80ef3ab0-7c57-4a42-8b90-3916c8ba8f6	Through Vegetable	167	95553
f0ff81e1-4141-8073-b4c1-0d01ca73c005	Company Dair	192	94990
030ff42-d5f3-4387-9e54-90a0e38702c	Western Mea	184	94157
57893364-12de-4c30-826e-4d91a0f6e8b	Pail Snack	175	94040
5e10e2db-4dad-4a0b-9967-d849793a268	Better Fruit	161	90966
62c08c9-61aa-4134-a546-24baf84b9dc	Early Fruit	163	90360
24b32b8b-9eef-4d22-8860-c9771b0c6d9	Intervien Dair	143	89170
9e00b8aa-e8ba-4fd1-b5fc-ebdb3fec64b9	Even Vegetable	160	88905
a861f68c-bb6d-47be-885f-51a8312949c	Total Dair	152	88764
2091292a-fef0-4390-8223-36e018b3abe	Drive Mea	174	88355
5ec10d67-2d4a-43f6-a472-062092ae814a	Than Vegetable	152	86620
ef8b3541-8e16-4339-a9ce-d6b6ed20ce2	Together Vegetable	149	86300
b8b7e4eb-0e4e-4286-b02e-c0d83924d44	Always Snack	148	86232
a857f89a-2041-4789-aabb-d7929230b94	Early Mea	139	86195
e450d338-edbe-489f-999e-801517f9f69	So Snack	162	85831
8aba364b-dffa-4752-9f89-c4498728919a	Field Fruit	146	85640
0628999d-2f51-4eaa-ab5e-7e54a383ca44	Late Vegetable	160	85583
6372a6d6-ec0d-4958-a054-7519c5314c39	Challenge Baker	157	85377
4a7a6a04-0ac7-a7a7-a7a7-a7a7-a7a7-a7a7	Weak Snack	144	84807



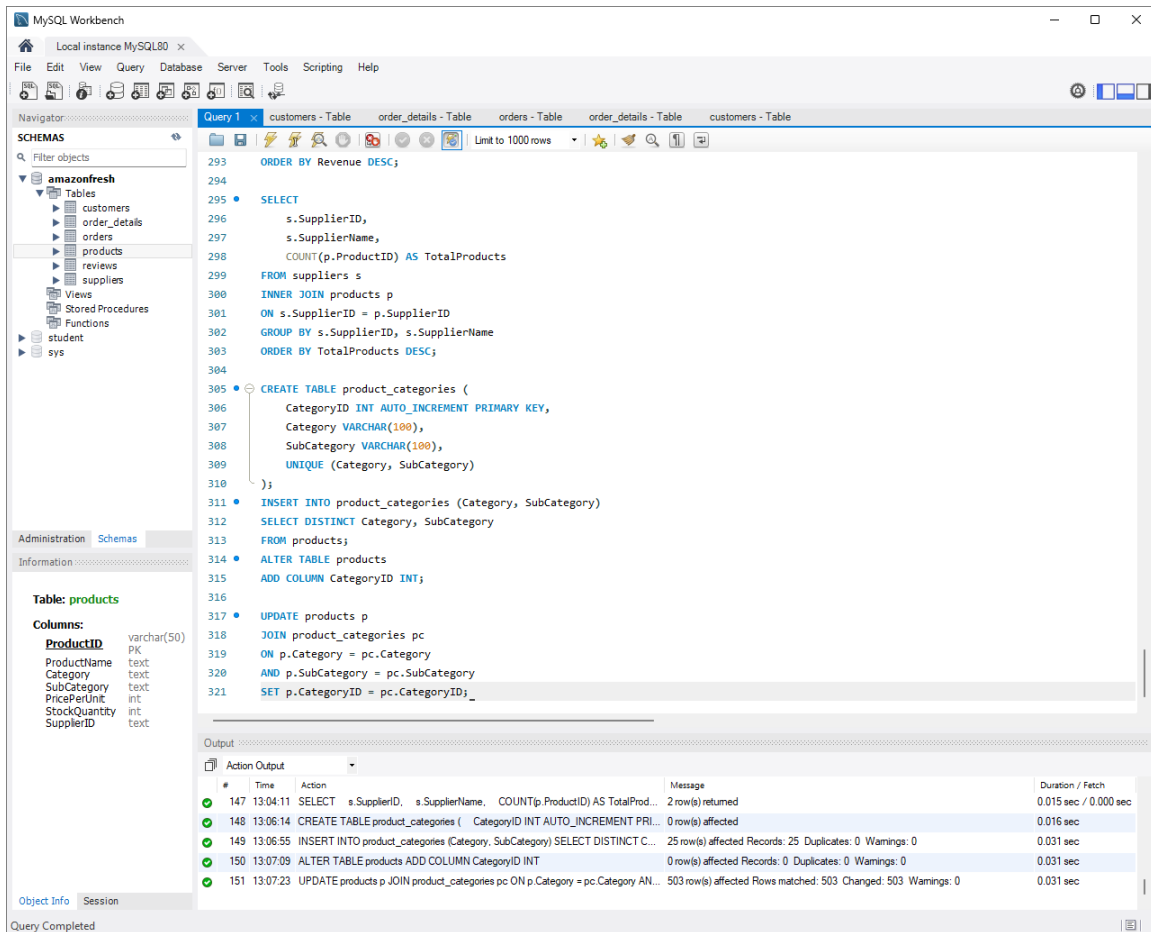
Task 12 - Normalization

The database was normalized to Third Normal Form (3NF).

A separate Product Categories table was created to remove redundancy and improve data organization.

Benefits:

- Reduced duplication
- Improved consistency
- Easier maintenance



Task 13 - Subqueries

Subqueries were used for advanced data retrieval.

Analysis performed:

- Top 3 products by revenue
- Customers who have not placed orders

Subqueries help solve complex business questions efficiently.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

amazonfresh

Tables

- customers
- order_details
- orders
- products
- reviews
- supplies
- Views
- Stored Procedures
- Functions
- student
- sys

Administration Schemas

Information

Table: products

Columns:

- ProductID varchar(50)
- PK
- ProductName text
- Category text
- SubCategory text
- PricePerUnit int
- StockQuantity int
- SupplierID text

Query 1

```

333 JOIN order_details od
334 ON p.ProductID = od.ProductID
335 GROUP BY p.ProductID, p.ProductName
336 ) AS product_sales
337 ORDER BY TotalRevenue DESC
338 LIMIT 3)
339 SELECT
340 CustomerID,
341 Name,
342 City

```

Result Grid

CustomerID	Name	City
024ebc32-77cc-4fca-b151-937425b5ab18	Erica Huber	New Carrie
02ce7230-4050-4d4a-b02d-eac1d174cd8a	William Cummings	Carlemouth
0439fc2f-e186-46a8-8b2f-7c77d4b6d323	Daniel Nguyen	Lavionshire
149253a3-3204-4052-a271-b4b902309bc3	Melinda McLaughlin	Haleton
1f7ab2eb-e720-4b02-885c-f4235198cc40	Omar Taylor	Port Tracy
30da8a58-1d78-4309-9fa2-8339bf7204b6	Sandra Martin	Olenburgh
34683326-95cf-4d41-wa9d-bf2b48f75014	Gloria Dominguez	Duenemouth
44189667-cf70-4b7c-8315-499f872a588e	Jeremy Duran	South Ambertown
424775d1-0bae-4aa3-bd69-95430a2b4d12	Kimberly Mclean	North Faithmouth
5861729a-9e27-49e8-87c4-68097932a5dc	Harrold Meyer	New Kaylemouth
ef3ec0cb-90ac-4c08-8756-a9bbf325222a	Joel Shaffer	South Lisland

Object Info Session customers 49

Query Completed

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

amazonfresh

Tables

- customers
- order_details
- orders
- products
- reviews
- supplies
- Views
- Stored Procedures
- Functions
- student
- sys

Administration Schemas

Information

Table: products

Columns:

- ProductID varchar(50)
- PK
- ProductName text
- Category text
- SubCategory text
- PricePerUnit int
- StockQuantity int
- SupplierID text

Query 1

```

317 UPDATE products p
318 JOIN product_categories pc
319 ON p.Category = pc.Category
320 AND p.SubCategory = pc.SubCategory
321 SET p.CategoryID = pc.CategoryID)
322
323 SELECT

```

Result Grid

ProductID	ProductName	TotalRevenue
1034fb7-fdce-49e2-9230-98af72d7fa15	Capital Snack	110875
6026d138-47e1-4052-a2df-8eab924ac57a	Society Vegetable	103430
227f5964-63b2-4923-9a65-73ac9e7e8b39	Fish Dair	100727

Result 48

Output

Action Output

Time	Action	Message	Duration / Fetch
148 13.06.14	CREATE TABLE product_categories (CategoryID INT AUTO_INCREMENT PRI...	0 row(s) affected	0.016 sec
149 13.06.15	INSERT INTO product_categories (Category, SubCategory) SELECT DISTINCT C...	25 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
150 13.07.09	ALTER TABLE products ADD COLUMN CategoryID INT	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
151 13.07.23	UPDATE products p JOIN product_categories pc ON p.Category = pc.Category AN...	503 row(s) affected Rows matched: 503 Changed: 503 Warnings: 0	0.031 sec
152 13.08.37	SELECT ProductID, ProductName, TotalRevenue FROM (SELECT ...	3 row(s) returned	0.047 sec / 0.000 sec

Object Info Session

Query Completed

Task 14 - Business Analysis

Business insights were generated from the database.

Analysis included:

- Cities with highest Prime membership concentration
- Top ordered product categories
- Product sales trends
- Customer purchasing behavior

These insights can support business decision-making and strategic planning.

The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'SCHEMAS' tree with 'amazonfresh' selected, containing tables like 'customers', 'order_details', 'orders', 'products', 'reviews', 'supplies', 'Views', 'Stored Procedures', 'Functions', 'student', and 'sys'. The main query editor shows a SQL query (lines 356-364) that selects product categories and their total ordered quantities, ordered by total ordered quantity in descending order, limited to 3 rows.

```
356 SELECT
357     p.Category,
358     SUM(od.Quantity) AS TotalOrdered
359 FROM products p
360 JOIN order_details od
361 ON p.ProductID = od.ProductID
362 GROUP BY p.Category
363 ORDER BY TotalOrdered DESC
364 LIMIT 3;
```

The 'Result Grid' shows the following data:

Category	TotalOrdered
Meat	11108
Fruits	9921
Snacks	9651

The bottom left pane shows the 'Table: products' structure:

Table: products

Columns:

- ProductID: varchar(50), PK
- ProductName: text
- Category: text
- SubCategory: text
- PricePerUnit: int
- StockQuantity: int
- SupplierID: text

The status bar at the bottom indicates 'Query Completed'.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

amazonfresh

- Tables
 - customers
 - order_details
 - orders
 - products
 - reviews
 - supplies
- Views
- Stored Procedures
- Functions
- student
- sys

Query 1 x customers - Table order_details - Table orders - Table order_details - Table customers - Table

Limit to 1000 rows

```
340 CustomerID,
341 Name,
342 City
343 FROM customers
344 WHERE CustomerID NOT IN (
345     SELECT CustomerID
346     FROM orders
347 );
348 SELECT
349     City,
```

Result Grid

Filter Rows: Exports: Wrap Cell Content: I

City	PrimeMemberCount
Kimberlyport	2
East Kristine	2
New David	2
Josephport	2
East Melissaschester	2
Devisfort	2
Floresmouth	2
Sarahmouth	2
Kristermouth	2
Robertbury	2
Port Denise	2
Martnshire	2
Whitneybury	2
South Michael	2
Port Christopher	2
North Lisa	2
Kellyport	2
Meganfort	1
South Georgechester	1
Castillostad	1
Bettyport	1
Baileyland	1
Lake Harold	1
Carlamouth	1
Lake Allen	1
North Benjaminview	1
Lake Kennethboro...	1
Lucasfurt	1
Nicolehaven	1
Port Craig	1
Codyhaven	1
Arthurmouth	1

Table: products

Columns:

Column	Type
ProductID	varchar(50)
ProductID	PK
ProductName	text
Category	text
SubCategory	text
PricePerUnit	int
StockQuantity	int
SupplierID	text

Object Info Session Result 50 x Read Only

Query Completed

Conclusion

The Amazon Fresh Analytics project successfully demonstrated the practical implementation of database concepts using MySQL. The project covered database design, relationships, constraints, normalization, data manipulation, joins, subqueries, and business analysis.

Through this project, valuable insights were obtained regarding customer behavior, product performance, supplier management, and business operations. The project strengthened understanding of SQL concepts and their real-world applications in data management and analytics.

PASTE SCREENSHOT HERE